

ASSIGNMENT- 06

* Objective :-

- 1) To calculate Result (Pass / Fail) based on marks.
- 2) To assign grades using Conditional formulas in Microsoft Excel.
- 3) To understand & apply the IF & Nested IF func.

* Procedure :-

- 1) (a) Enter Student Name in (column A) & Marks in Column B, & result in column C2.
- (b) Use = IF (B2 > 50; "PASS", "FAIL")

- 2) (a) Click on cell D2 (Grades Column)
- (b) Use = IF (B2 >= 80, "A")
 IF (B2 >= 60, "B")
 IF (B2 >= 40, "C", "FAIL"))



Assignment - 07

INSTRUCTIONS :-

Task 1: Calculate the no. of complete years of service for each employee.

Formula Example: = DATE DIF (C2, D2, "Y")

Task 2: Calculate the no. of remaining months (ignoring years).

Formula Example: = DATEDIF (C2, D2, "YM")

Task 3: Calculate the no. of ascending days (ignoring weeks & years).

Task 4: Create a column that shows full service period in the format:

Example: "10 years, 3 months, 12 Days".

Hint: Use CONCATENATE or TEXTJOIN with DATEDIF overall.

Task 5: Identify employees who have completed more than 10 years of service.

Hint: Use IF condition with DATEDIF in your column.

Task 6: Calculate how many employees joined before 2015.

Hint: Use COUNTIF with JOINING - DATE.

Task 7: Create a summary table showing:

→ No. of employees with < 5 yrs service.

→ No. of employees with 5-10 yrs service.

→ No. of employees with > 10 yrs service.

Teacher's Signature

Employee ID	Employee Name	Joining Date	Current Date	Years	Months	Days	Full Service Period	More Than 10 Yrs
E101	John Doe	2018-12-01	2023-09-30	4	9	27	4 Years, 9 Months, 27 Days	No
E102	Jane Smith	2020-03-20	2023-09-30	3	6	10	3 Years, 6 Months, 10 Days	No
E103	David Brown	2015-05-10	2023-09-30	8	4	20	8 Years, 4 Months, 20 Days	No
E104	Emily White	2018-11-01	2023-09-30	4	10	29	4 Years, 10 Months, 29 Days	No
E105	Michael Green	2017-02-15	2023-09-30	6	7	15	6 Years, 7 Months, 15 Days	No
E106	Sarah Lee	2019-08-01	2023-09-30	4	1	29	4 Years, 1 Month, 29 Days	No
E107	Robert King	2016-01-01	2023-09-30	7	8	29	7 Years, 8 Months, 29 Days	No
E108	Laura Hall	2021-03-01	2023-09-30	2	6	29	2 Years, 6 Months, 29 Days	No
E109	James Taylor	2020-05-24	2023-09-30	3	4	6	3 Years, 4 Months, 6 Days	No
E110	Anna Wilson	2017-01-01	2023-09-30	6	8	29	6 Years, 8 Months, 29 Days	No
E111	Thomas Moore	2022-04-01	2023-09-30	1	5	29	1 Year, 5 Months, 29 Days	No
E112	Patricia King	2023-01-01	2023-09-30	0	8	29	8 Months, 29 Days	No
E113	Richard Baker	2021-06-01	2023-09-30	2	3	29	2 Years, 3 Months, 29 Days	No
E114	Olivia Clark	2022-12-15	2023-09-30	0	9	15	9 Months, 15 Days	No
E115	Daniel Lewis	2020-10-10	2023-09-30	3	9	20	3 Years, 9 Months, 20 Days	No
E116	Grace Hall	2021-03-01	2023-09-30	2	6	29	2 Years, 6 Months, 29 Days	No
E117	Kevin Taylor	2022-02-27	2023-09-30	1	7	3	1 Year, 7 Months, 3 Days	No
E118	Michelle Adams	2019-07-19	2023-09-30	4	2	11	4 Years, 2 Months, 11 Days	No
E119	Christopher	2023-08-11	2023-09-30	0	1	19	1 Month, 19 Days	No
E120	Amanda Lee	2018-11-25	2023-09-30	4	8	25	4 Years, 8 Months, 25 Days	No

after 1990.

19. Highlight employees whose salary is above the average salary in light blue.

20. Highlight employees in Region - East & Salary < 9500 in red.

Step-by-step solution:-

1) Salary > 9500 (light green)

• Range: A2: I107

• Home → Conditional Formatting → New Rule → Use a formula to determine which cells to format.

• Formula: = \$I2 > 9500.

• Format: Fill = light green → OK.

2) Admin dept with Salary < 9500 (red)

• Range: A2: I107

• New Rule → Use a formula

• Formula: = AND (\$E2 = "Admin", \$I2 < 9500)

• Format: Fill = red → OK.

3) Salary Data Range

• Range: I2: I107

• Home → Conditional Formatting → Data Bars → (choose any Gradient or Solid Fill).

4) Region - East (yellow)

• Range: A2: I107

• New Rule → Use a formula

• Formula: = AND (\$F2 = "East")

• Format: Fill = yellow → OK.

5) 3-Color Scale on Salary

• Range: I2: I107

HR Data									
Emp ID	Emp Name	Region	Dept	Salary	Age	Gender	Marital	Education	Experience
1	John Doe	East	Admin	12000	35	M	Married	High School	10
2	Jane Smith	West	Admin	10000	30	F	Single	High School	5
3	Bob Johnson	East	Admin	8000	25	M	Single	High School	3
4	Alice Brown	West	Admin	6000	20	F	Single	High School	1
5	Charlie Davis	East	Admin	4000	15	M	Single	High School	0
6	Diana Evans	West	Admin	3000	10	F	Single	High School	0
7	Frank Green	East	Admin	2000	5	M	Single	High School	0
8	Grace Hill	West	Admin	1500	3	F	Single	High School	0
9	Henry King	East	Admin	1000	2	M	Single	High School	0
10	Ivy Lee	West	Admin	500	1	F	Single	High School	0
11	Jack Miller	East	Admin	12000	35	M	Married	High School	10
12	Jill Wilson	West	Admin	10000	30	F	Single	High School	5
13	Kevin Moore	East	Admin	8000	25	M	Single	High School	3
14	Laura Taylor	West	Admin	6000	20	F	Single	High School	1
15	Mark Anderson	East	Admin	4000	15	M	Single	High School	0
16	Nancy White	West	Admin	3000	10	F	Single	High School	0
17	Oscar Black	East	Admin	2000	5	M	Single	High School	0
18	Peter Gray	West	Admin	1500	3	F	Single	High School	0
19	Quinn Brown	East	Admin	1000	2	M	Single	High School	0
20	Rachel Green	West	Admin	500	1	F	Single	High School	0
21	Samuel White	East	Admin	12000	35	M	Married	High School	10
22	Tina Black	West	Admin	10000	30	F	Single	High School	5
23	Uma Gray	East	Admin	8000	25	M	Single	High School	3
24	Victor Brown	West	Admin	6000	20	F	Single	High School	1
25	Wendy Green	East	Admin	4000	15	M	Single	High School	0
26	Xavier White	West	Admin	3000	10	F	Single	High School	0
27	Yara Black	East	Admin	2000	5	M	Single	High School	0
28	Zoe Gray	West	Admin	1500	3	F	Single	High School	0
29	Adam Brown	East	Admin	1000	2	M	Single	High School	0
30	Eve Green	West	Admin	500	1	F	Single	High School	0

- Hows → Conditional Formatting → Color Scales → 3-Color Scale (green - Yellow - Red).
(You can customize min/max under "Manage Rules" if needed)
- 8) First Name starts with "S"
Range: A2 : I {N}
New Rule → Use a formula
Formula: = LEFT (UPPER (\$A2), 1) = "S"
Format: choose a Highlight (eg. light blue) → OK.
4) Hire date before 01-Jan-1990.
Range: A2 : I {N}
New Rule → Use a formula
Formula: = \$H2 < DATE (1990, 1, 1)
Format: choose a Highlight (eg. light orange) → OK.
8) COO dept with salary exactly 8000.
Range: A2 : I {N}
New Rule → Use a formula
Formula: choose a distinct fill (eg. teal) → OK.
• Formula: = AND (\$E2 = "COO", \$I2 = 8000)
9) Top 3 Highest salaries (blue)
Range: I2 : I {N}
• Hows → Conditional Formatting → Top/Bottom Rules → Top 10 items ... → Change 10 to 3.
• Format: fill = blue (or any) → OK.
10) Bottom 3 salaries (orange)
Range: I2 : I {N}
• Hows → Conditional Formatting → Top/Bottom Rules → Bottom 10 items ... → Change 10 to 3.
• Format: fill = orange → OK.

11) Dept code = 70 (pink)

• Range : A2 : I {N}

• New Rule $\rightarrow$ Use a formula

• formula : = \$G2 = 70

• format : fill = pink $\rightarrow$ OK.

12) seen sets for salary (traffic lights)

• Range : I2 : I {N}

• Home $\rightarrow$ conditional formatting $\rightarrow$ seen sets $\rightarrow$ choose a 3-ion

set (eg. 3 traffic lights).

(Optional : Manage Rules $\rightarrow$ Edit Rule to set "Seen seen

Only" or adjust threshold)

13) field + last name length > 12 character

• Range : A2 : I {N}

• New Rule $\rightarrow$ Use a formula

• formula : = LEN (\$G2 & \$C2) > 12

• format : choose a lightlight $\rightarrow$ OK.

14) True Rules :

a) Male with salary > 9500 (green)

• Range : A2 : I {N}

• New Rule $\rightarrow$ Use a formula

• formula : = AND (\$D2 = "Male", \$I2 > 9500)

• format : fill = green $\rightarrow$ OK.

b) Female with salary < 9500 (red)

• Range : A2 : I {N}

• New Rule $\rightarrow$ Use a formula

• formula : = AND (\$D2 = "Female", \$I2 < 9500)

• format : fill = red $\rightarrow$ OK.

(Even the rule order doesn't conflict; they're independent).

- 15) Here year is 1988
- Range: A2: I3:A3
 - Need Rule $\rightarrow$ Use a formula
 - Formula: = YEAR (\$A2) = 1988
 - Format: Choose a lightlight $\rightarrow$ OK
- 16) but name contain "a" (case-insensitive)
- Range: A2: I3:A3
 - Need Rule $\rightarrow$ Use a formula
 - Formula: = ISNUMBER (SEARCH ("a" , \$C2))
 - Format: Choose a lightlight $\rightarrow$ OK
- 17) Alternate row shading (light grey)
- Range: A2: I3:A3
 - Need Rule $\rightarrow$ Use a formula
 - Formula: = ISODD (ROW ())
 - Format: Fill: light grey $\rightarrow$ OK
- (If you want shading to start from row 2 specifically regardless of sheet row number, you can also use: MOD (ROW () - ROW (\$A\$2) , 2) = 0)
- 18) Admin dept hired after 1990.
- Range: A2: I3:A3
 - Need Rule $\rightarrow$ Use a formula
 - Formula: = AND (\$C2 < "Admin" , \$H2 > DATE (1990 , 1 , 1))
 - Format: Choose a lightlight $\rightarrow$ OK
- 19) Salary above overall average (light blue)
- Range: A2: I3:A3
 - Need Rule $\rightarrow$ Use a formula
 - Formula: = \$I2 > AVERAGE (\$I : \$I)
 - Format: Fill: light blue $\rightarrow$ OK

20) Region = east of colony < 9500 (red)

- Range : A2 : I2 : I2
- New Rule → Use a formula
- Formula : = AND (lower (I2 : E2) = "east", I2 < 9500)
- Format : fill = red → OK.

ASSIGNMENT - 09

Q1 Find the Product Name for Product ID P005.

→ Answer

Formula: VLOOKUP("P005", B2:D16, 2, FALSE)

Q2 Retrieve the Product Price of Tablet (P015).

→ Answer

Formula: ("P015", B2:D16, 3, FALSE)

Q3 Which product ID corresponds to the price of ₹18,000?

→ Answer (Smart watch)

Formula: VLOOKUP(18000, D2:D16, 1, FALSE)

Q4 Find the product price for Headphones (P015).

→ Answer

Formula: VLOOKUP("P015", B2:D16, 3, FALSE)

Q5 What will VLOOKUP return if you search for P999?

→ Answer

Formula: VLOOKUP("P999", B2:D16, 2, FALSE)

Q6 Find the Product Name of the item priced at ₹25,000.

→ Answer (Smartphone)

Formula: VLOOKUP(25000, D2:D16, 2, FALSE)

Q7 Using Product ID P007, find the Product Price.

→ Answer

Formula: VLOOKUP("P007", B2:D16, 3, FALSE)

Q8 Get the Product ID of Keyboard.

→ Answer

Formula: VLOOKUP("Keyboard", C1:D16, 1, FALSE)

Q9 Find the Product Name for Product ID P002.

→ Answer

Formula: VLOOKUP("P002", B2:D16, 2, FALSE).

S. No.	Product ID	Product Name	Product Price
1	P001	Laptop	55000
2	P002	Mouse	800
3	P003	Keyboard	1200
4	P004	Monitor	9000
5	P005	Printer	7000
6	P006	Scanner	6000
7	P007	Webcam	2500
8	P008	Headphones	1500
9	P009	Speaker	3500
10	P010	Microphone	2000
11	P011	External HDD	4500
12	P012	USB Drive	700
13	P013	Smartphone	30000
14	P014	Tablet	20000
15	P015	Smartwatch	12000
16	P016	Charger	1500
17	P017	Router	4000
18	P018	Projector	25000
19	P019	Graphics Card	18000
20	P020	RAM Module	6000

Q10. Retrieve the product price for product ID P014.
→
12000

Formula: `VLOOKUP("P014", B2:D16, 3, FALSE)`.

Q11. What will `VLOOKUP` return if column index > no. of columns?
→
#REF!

Example formula: `VLOOKUP("P003", B2:D16, 5, FALSE)`

Q12. Retrieve the product name for product ID P008.
→
Q'sawano

Formula: `VLOOKUP("P008", B2:D16, 2, FALSE)`.

ASSIGNMENT-10

PART-A : Conceptual questions

Q1. What is the difference b/w INDEX & VLOOKUP?

→ INDEX returns a value from a specific position, while VLOOKUP searches vertically & returns a value from another column.

Q2. Explain how the MATCH func is used with INDEX.

→ MATCH finds the position of a value, which INDEX uses to return the corresponding unit.

Q3. Why is INDEX + MATCH preferred over VLOOKUP in many cases?

→ INDEX + MATCH is preferred because it is more flexible, faster & can look in any direction without breaking when columns change.

PART-B : Practical Exercises (Use Employee - Data Sheet)

Q1. Write an INDEX formula to return the name of the 2nd employee in the list.

→ INDEX (name Range, 2) → Ankit

Q2. Use INDEX to return the salary of the 5th employee.

→ INDEX (Salary Range, 5) → 48000

Q3. Use MATCH to find the position of 'Finance' in the Department column.

→ MATCH ('Finance', Department Range, 0) → 3

Q4. Use MATCH to find the position of employee 10104 in the Employee ID column.

Employee ID	Name	Department	Salary
101	Rahul	IT	50000
102	Priya	HR	45000
103	Ankit	Finance	55000
104	Neha	IT	52000
105	Suresh	Marketing	48000

Product ID	Product Name	Price	Stock
P001	Laptop	65000	12
P002	Mobile	25000	30
P003	Tablet	18000	20
P004	Headphones	2500	50
P005	Monitor	12000	15

→ MATCH (104, Employee ID Range, 0) → 14

Q2a) Write a formula to find the salary of 'Binge' using INDEX + MATCH.

→ INDEX (Salary Range, MATCH ('Binge', Name Range, 0)) → 45000.

Q2b) Find the Department of 'Suresh' using INDEX + MATCH.

→ INDEX (Department Range, MATCH ('Suresh', Name Range, 0)) → Marketing.

Q2c) Create a search box where user types Employee name, returns salary using INDEX + MATCH.

→ INDEX (Salary Range, MATCH (Search Name, Name Range, 0))

Challenge of INDEX (Stock Range, MATCH ('Tablet', Product Name Range, 0)) → 20

Challenge to INDEX (Price Range, MATCH (Input ID, Product ID Range, 0)).

PART-C : Challenge Questions (Use Product - Data Sheet)

Q1) Write a formula using INDEX + MATCH to find the stock of 'Tablet'.

→ =INDEX (D2:D10, MATCH ("Tablet", B2:B10, 0))

Q2) Create a dynamic lookup: if user types a Product ID, returns its price.

→ =INDEX (C2:C10, MATCH (F1, A2:A10, 0))